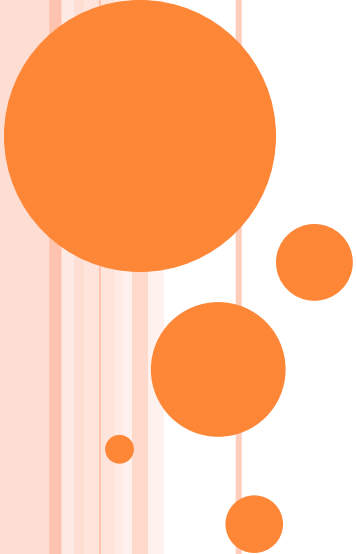




# **INFORMATION SYSTEM LIFE CYCLE**

**General Information about the Information  
System Life Cycle**

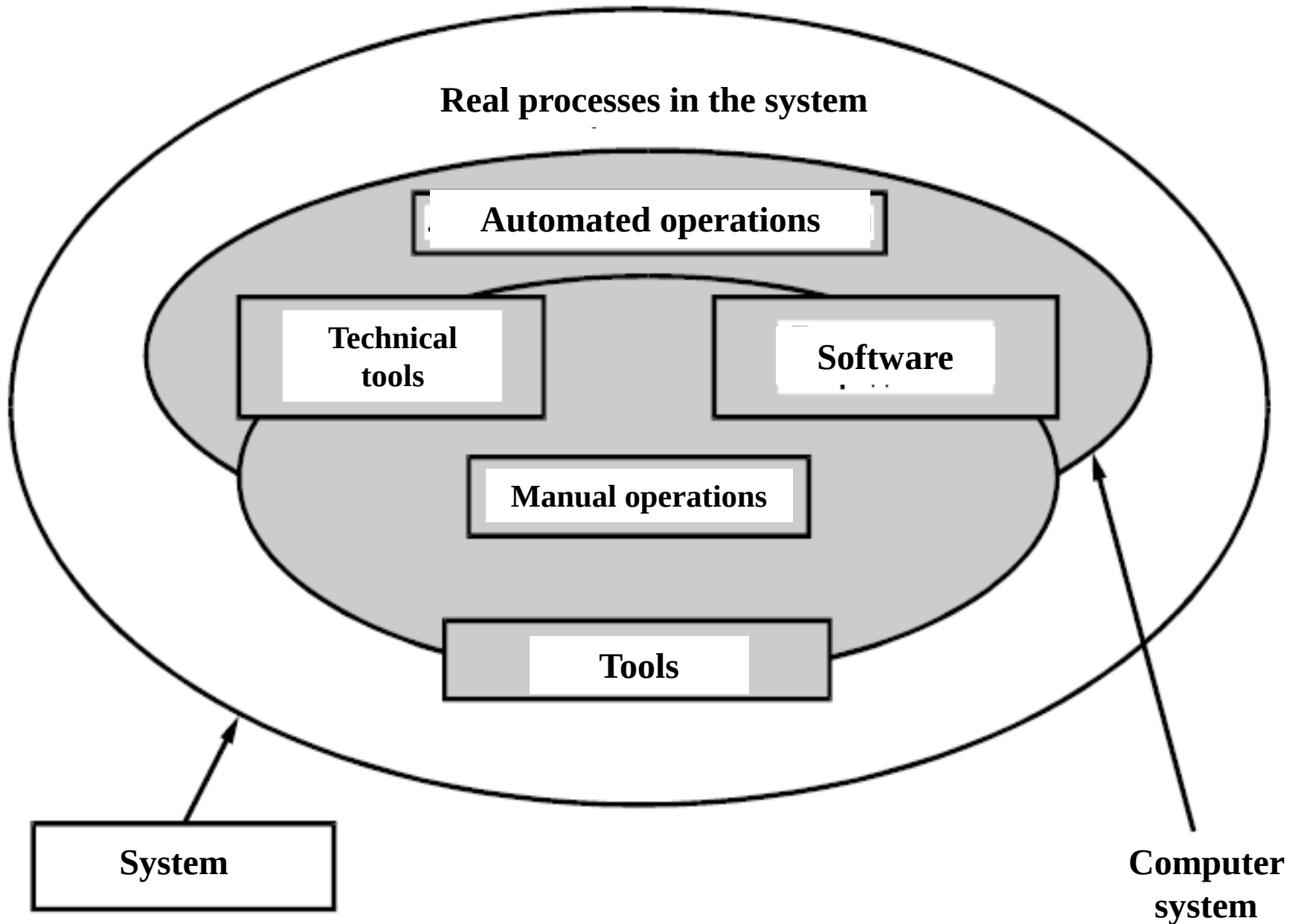
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# WHAT IS A SYSTEM?

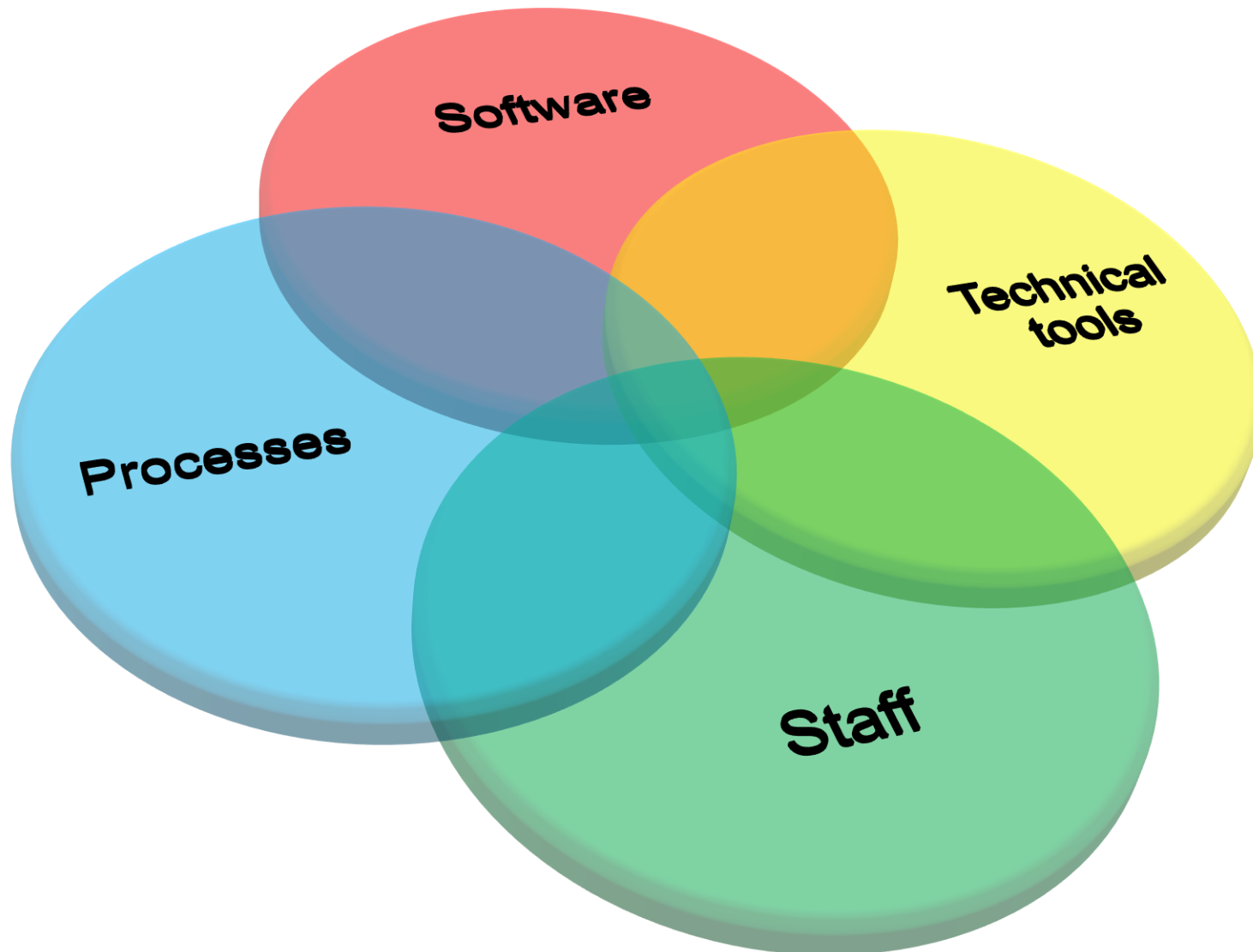
**An information system is a complex consisting of automated processes, hardware and software tools, devices, and personnel, capable of meeting specified needs or objectives**



# INFORMATION SYSTEM



# AN INFORMATION SYSTEM IS:

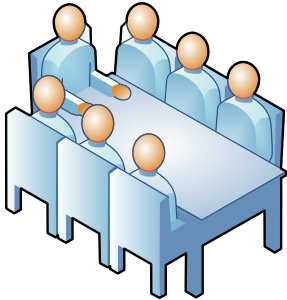


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# WHERE DOES THE LIFE OF AN INFORMATION SYSTEM BEGIN?



- **There is a Business that is functioning normally. However, at some point it sets goals whose achievement requires the use of information technologies**



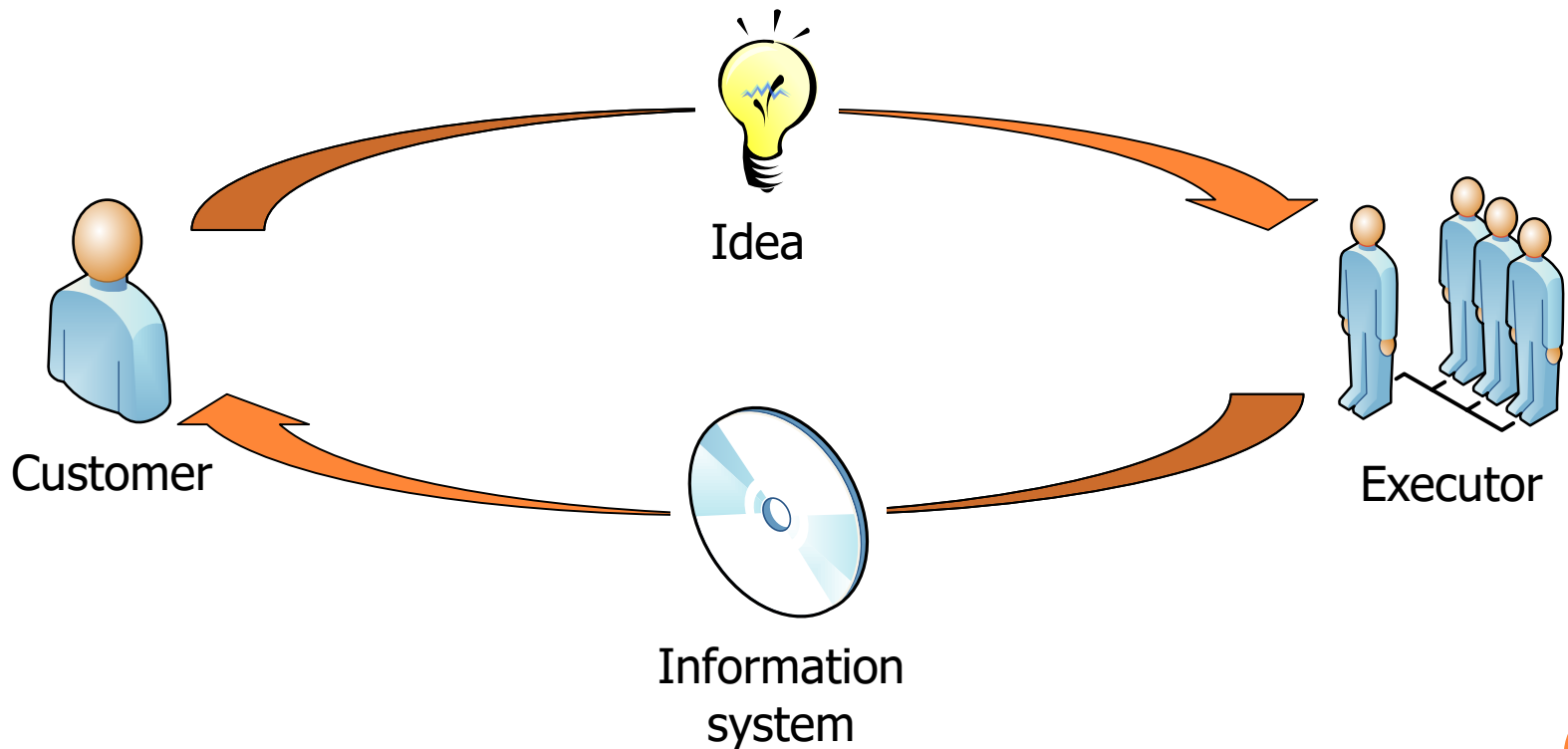
- **The Business begins to develop an understanding of how and which information technologies can help it achieve its goals**



- **As a result, the Business arrives at an idea expressed in the form of expectations, wishes, and so on**

# CUSTOMER AND EXECUTOR

When the Business decides to implement its idea, it begins looking for an Executor. The Business itself acts as the Customer.



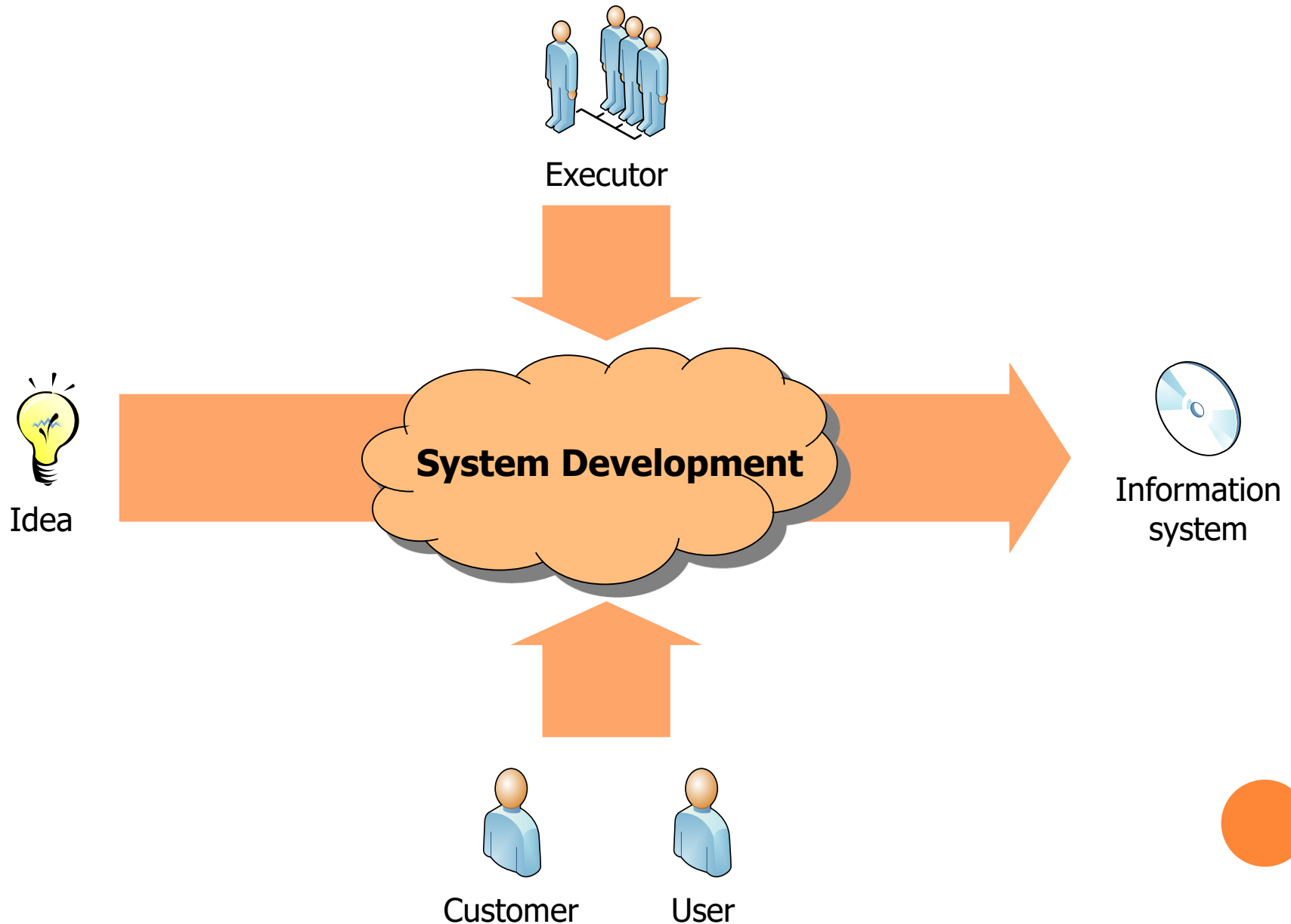
# WHAT DOES THE CUSTOMER DO?

The Customer acts as a consultant to the Executor on matters related to the Customer's wishes and expectations

The Customer monitors the Executor's compliance with its obligations

The Customer monitors the quality of the system being created and its compliance with the requirements

# WHAT DOES THE EXECUTOR DO?



# WHAT ABOUT THE USER?

The system Customer is the person who pays for its implementation. In general, the Customer may never even see the system or have a clear understanding of how it will be used in the Business

Users are the people who will use the system in their daily activities. Therefore, they can say what the system should do and how it should work so that, ultimately, the goals that prompted the Business to create it are achieved

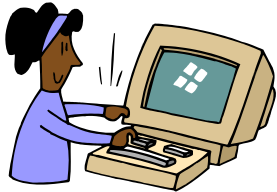
In addition, Users possess information about how the Business currently operates and what problems it has

Therefore, Users are very important participants in system development

# WHAT IS MEANT BY SYSTEM DEVELOPMENT?



- **First, the Executor tries to understand what the Customer wants and what the system should do to satisfy the Customer. The Executor identifies the system requirements**



- **After the system requirements have been identified, the architecture of the system and its components is designed. This is followed by development (system programming)**



- **Before delivering the system to the Customer, it is necessary to verify that it works and meets the stated requirements. The system is tested**



- **When the system is ready, it is delivered to the Customer. The system is deployed and put into trial operation, followed by transition to normal operation**

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# DEFINITION OF THE LIFE CYCLE

*The life cycle (LC) of an information system is an ordered set of activities carried out and managed for the purpose of creating, implementing, and operating an information system*

The life cycle begins with the emergence of the system idea and ends with the system being taken out of operation and disposed of

# DOCUMENT REGULATING THE LIFE CYCLE

One of the documents that officially establishes the general life cycle structure of software is ISO 12207

The standard can be applied to the acquisition of systems and provision of related services, as well as to the supply, development, operation, and maintenance of systems

ISO 12207 defines a set of processes, activities, and tasks intended to be adapted to the conditions of specific software projects

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# INFORMATION SYSTEM LIFE CYCLE

## Primary Life Cycle Processes

System development,  
operation, and  
maintenance  
processes

## Supporting Processes

Documentation management, configuration management,  
quality management, risk management, resolution of  
emerging issues

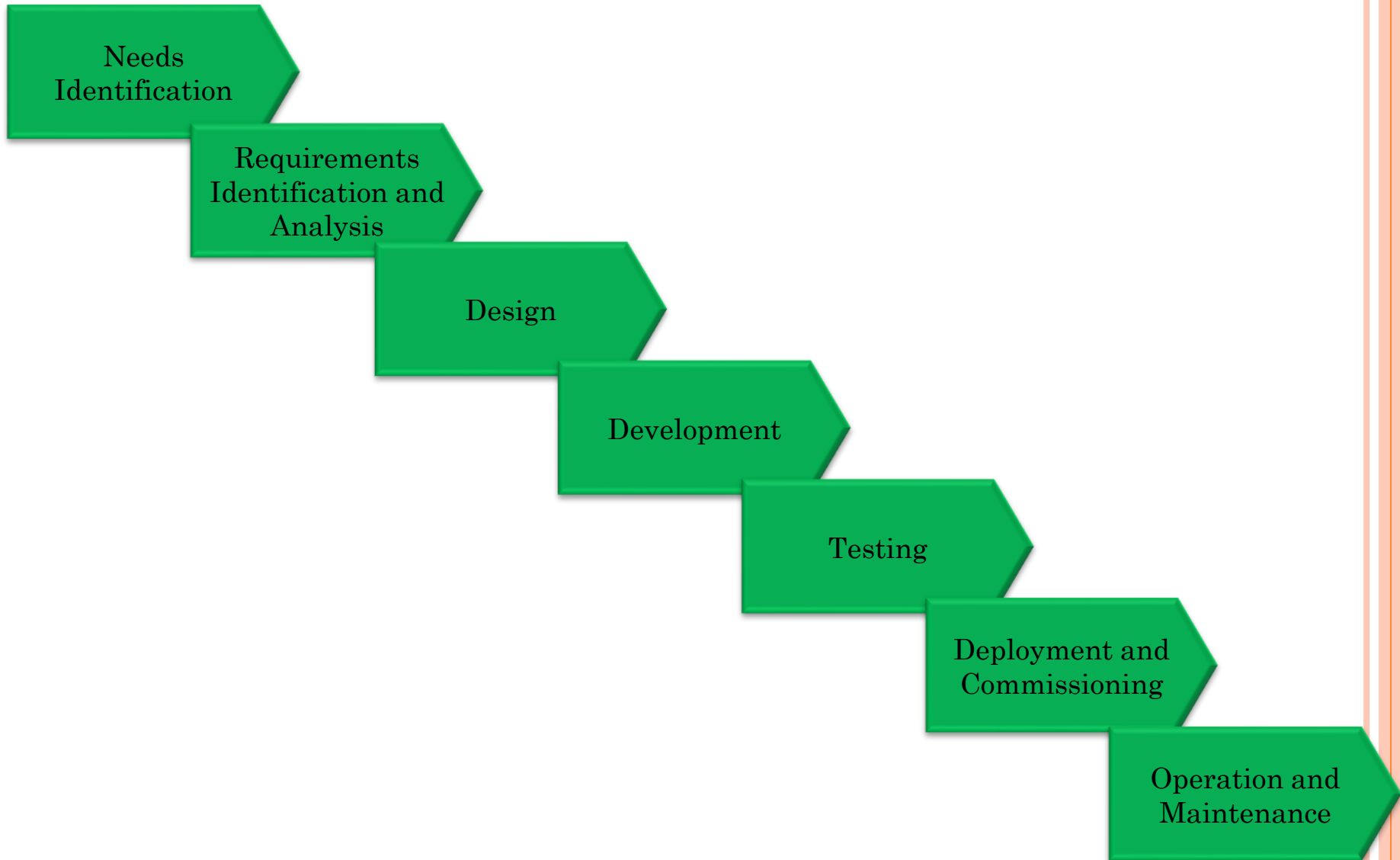
Ensure  
successful  
implementation  
of the primary  
processes

## Organizational Processes

Life cycle management, creation and maintenance of infrastructure, training  
of life cycle participants

Provide the  
infrastructure for  
carrying out life  
cycle processes

# LIFE CYCLE STAGES

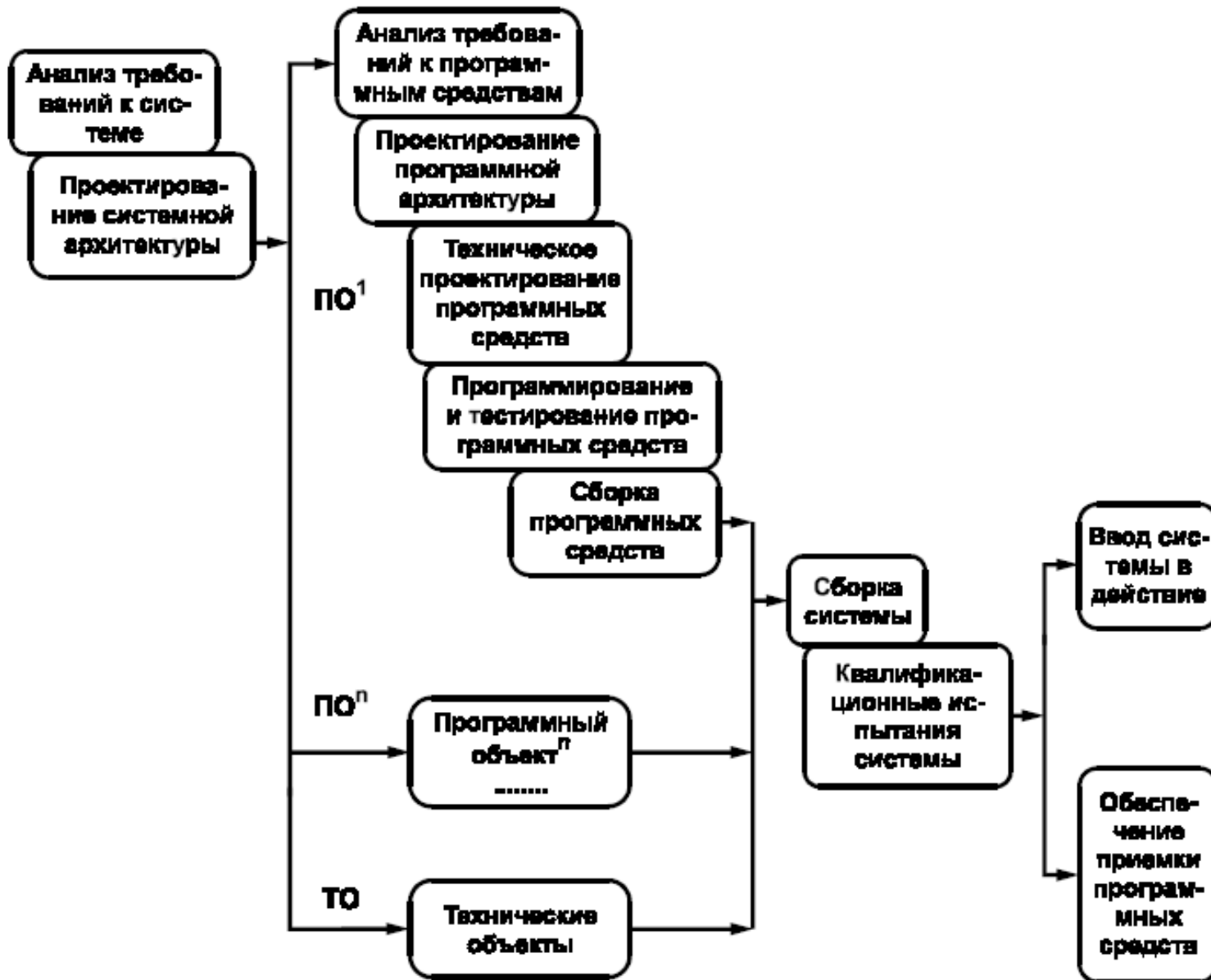


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# WATERFALL LIFE CYCLE MODEL

*The waterfall life cycle model implements the principle of performing each life cycle stage only once*

# WATERFALL LIFE CYCLE MODEL



# ADVANTAGES OF THE WATERFALL MODEL

- All system requirements and characteristics are defined once throughout the entire life cycle
- The entire system is implemented at once, and migration from legacy systems to the new system is performed only once

# DISADVANTAGES OF THE WATERFALL MODEL

The following disadvantages are inherent in this model, making its use highly risky under one or more of the following conditions:

- System requirements are not defined clearly enough
- The system is too large for all development work to be performed only once
- Rapid changes in system development technologies are expected
- Changes in system requirements are possible
- There are resource constraints, such as limited funding or personnel
- The Customer expects an intermediate product to already be usable

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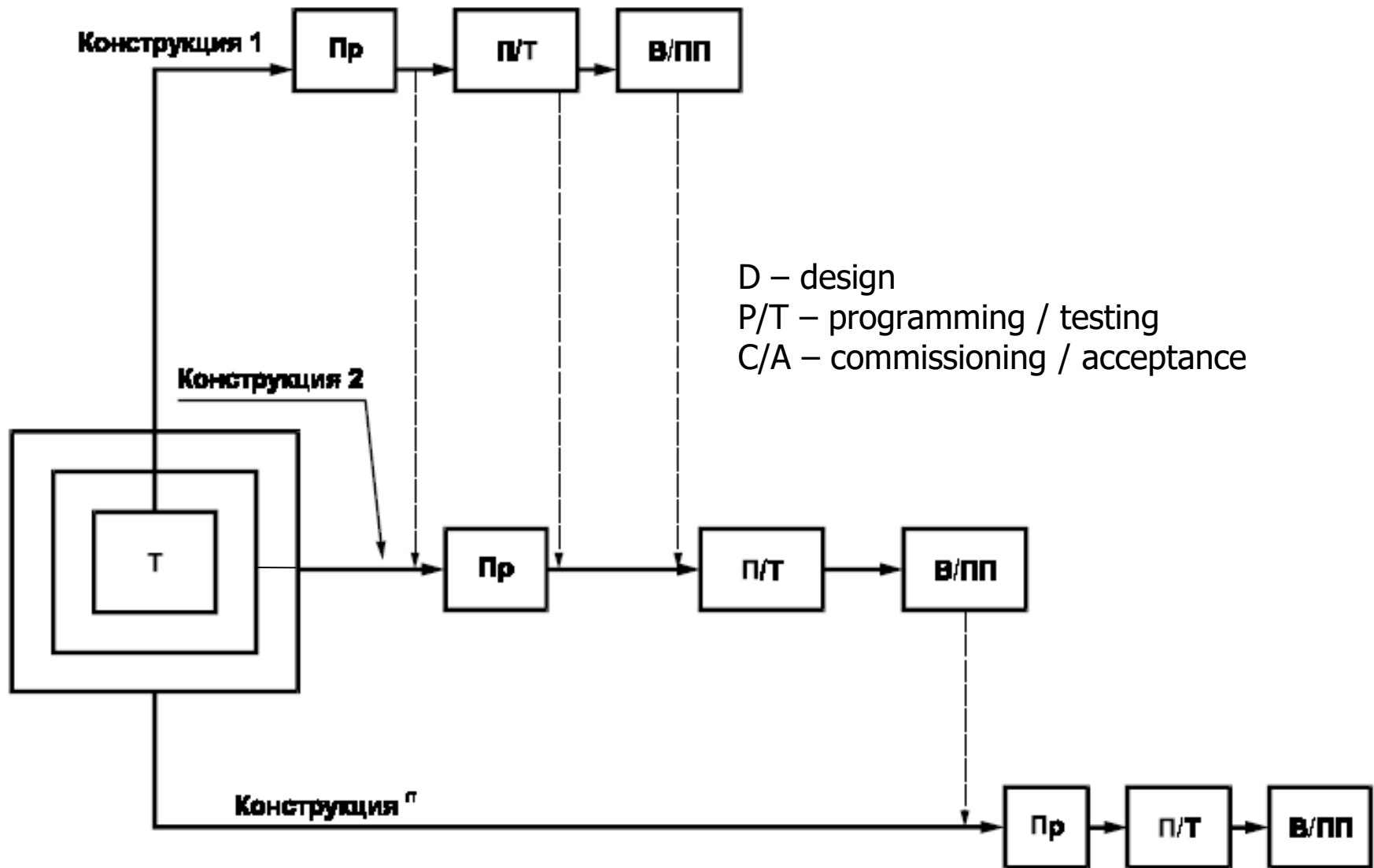
# INCREMENTAL LIFE CYCLE MODEL

The incremental life cycle model implements planned system enhancement

System development using the incremental model begins with a set of requirements and involves developing a sequence of builds.

A build is a fragment of the system that implements part of the requirements

# INCREMENTAL LIFE CYCLE MODEL



# ADVANTAGES OF THE INCREMENTAL MODEL

Advantages of the incremental model:

- The intermediate product is usable
- Natural division of the system into expandable components (increments)
- Ability to involve additional resources in system development

# DISADVANTAGES OF THE INCREMENTAL MODEL

Use of the incremental model is difficult in the following cases:

- System requirements are not defined clearly enough
- The Customer wants all system capabilities at once
- Rapid changes in work technologies are expected
- Changes in system requirements are possible
- The ability to engage resources (funding or personnel) for a long period is limited

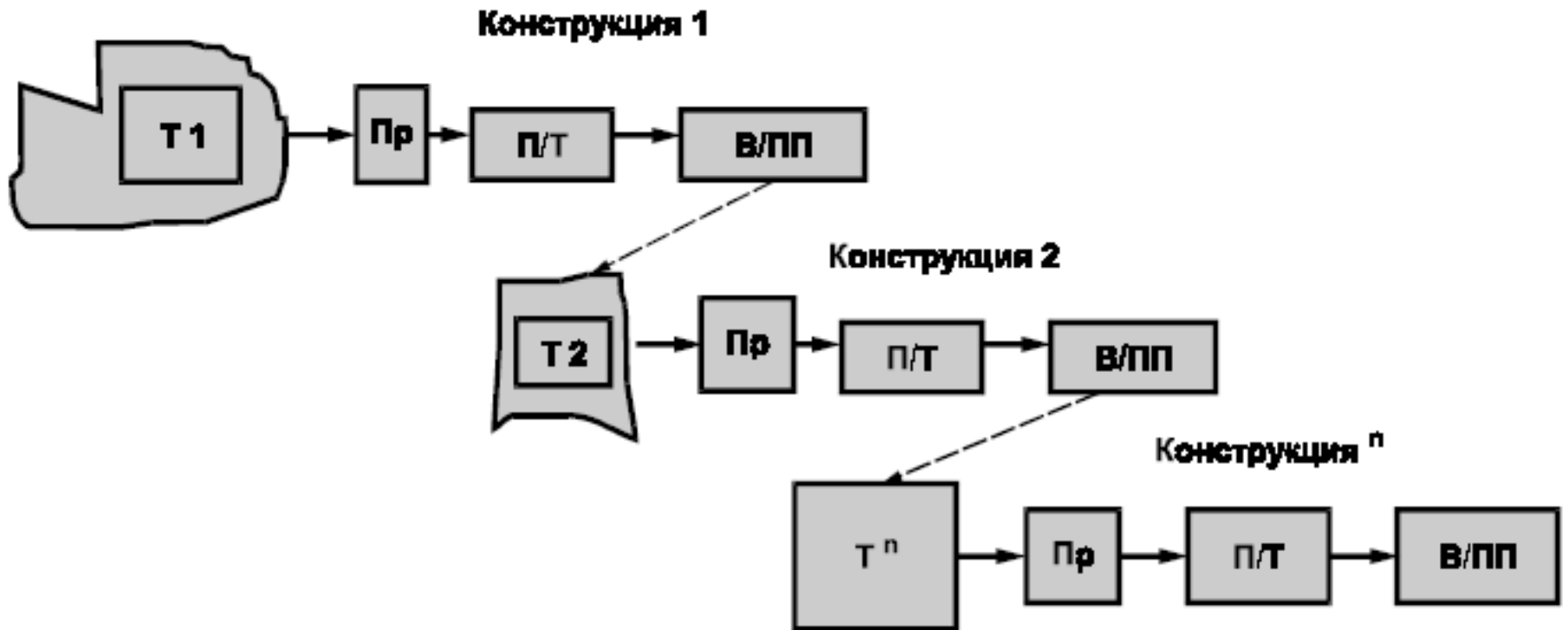
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# EVOLUTIONARY LIFE CYCLE MODEL

In the evolutionary life cycle model, the system is also developed as separate builds, but unlike the incremental model, the requirements cannot initially be fully understood and established

In this model, requirements are established partially and refined in each subsequent build

# EVOLUTIONARY LIFE CYCLE MODEL



# ADVANTAGES OF THE EVOLUTIONARY MODEL

Advantages of the evolutionary life cycle model:

- The intermediate product is usable
- Natural division of the system into expandable components (increments)
- Ability to engage personnel and resources as needed
- Necessary feedback from the User is available for a complete understanding of system requirements
- Monitoring of technology changes is simplified

# DISADVANTAGES OF THE EVOLUTIONARY LIFE CYCLE MODEL

Use of the evolutionary model is not advisable in the following cases:

- All system requirements are defined initially
- Limited ability to engage resources (funding or personnel) for a long period

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# CRITERIA FOR COMPARING DIFFERENT APPROACHES

Main criteria for comparing different approaches:

- Resources required to develop the system
- Duration of the system development project
- Completeness and clarity of system requirements at the beginning of design and development
- Usability of the intermediate development product
- Ability to change project requirements and work technology at different stages of the development process

# COMPARISON

| Criterion                      | Waterfall                                                | Incremental                                                            | Evolutionary                                                           |
|--------------------------------|----------------------------------------------------------|------------------------------------------------------------------------|------------------------------------------------------------------------|
| Resource Engagement            | Many resources for a relatively short period             | Resources engaged for a long period, with flexible resource management | Resources engaged for a long period, with flexible resource management |
| Duration                       | Short duration due to full parallelization of activities | Long duration due to sequential design and implementation of builds    | Long duration due to sequential design and implementation of builds    |
| Requirements Completeness      | Required                                                 | Required                                                               | Not required                                                           |
| Intermediate Product Usability | No                                                       | Yes                                                                    | Yes                                                                    |
| Ability to Change Requirements | No                                                       | No                                                                     | Available                                                              |

# TASK

- 1) Which life cycle model is used in your project, and for what reasons?*
- 2) Which life cycle model would you like to work with, and why?*